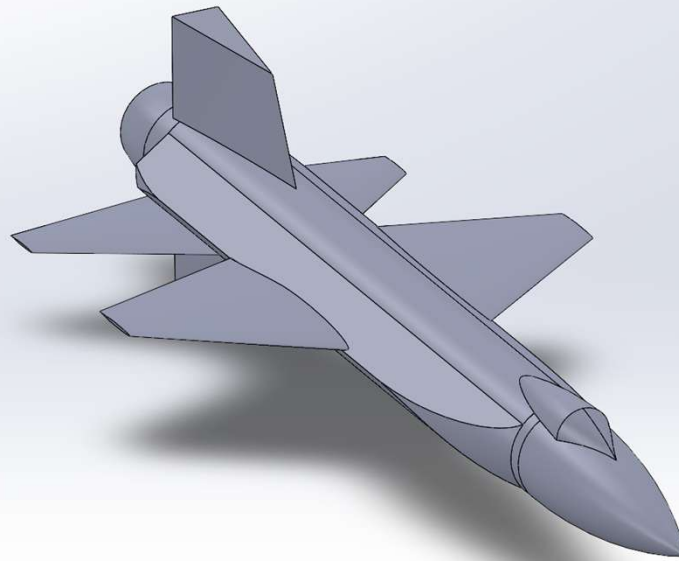
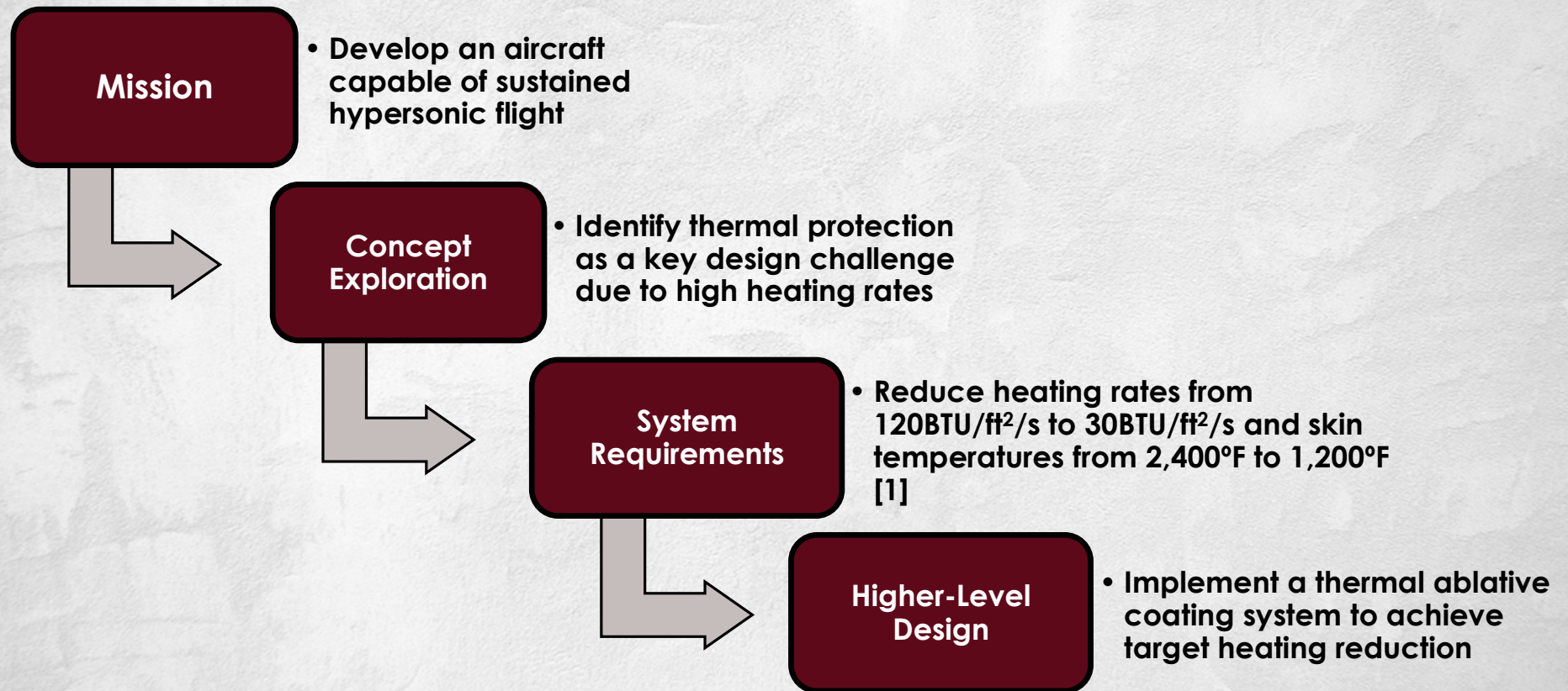


X-15 Final Briefing



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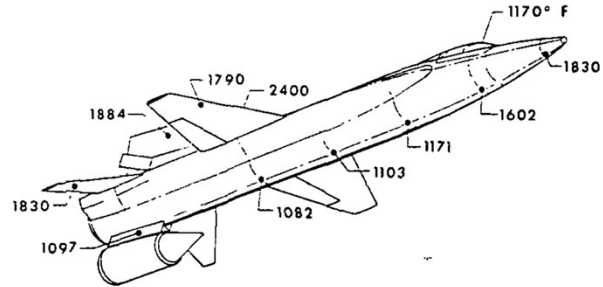
Systems V-Diagram



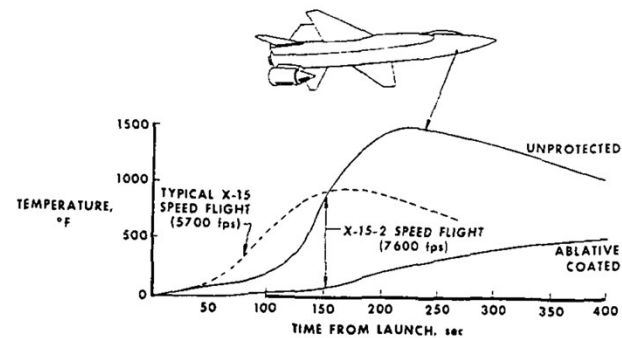
Simulation Overview

CALCULATED TEMPERATURES ON UNPROTECTED X-15-2 AIRFRAME

MAXIMUM VELOCITY = 8,000 fps
ALTITUDE = 100,000 ft



CALCULATED X-15-2 TEMPERATURES



[2]

SOLIDWORKS

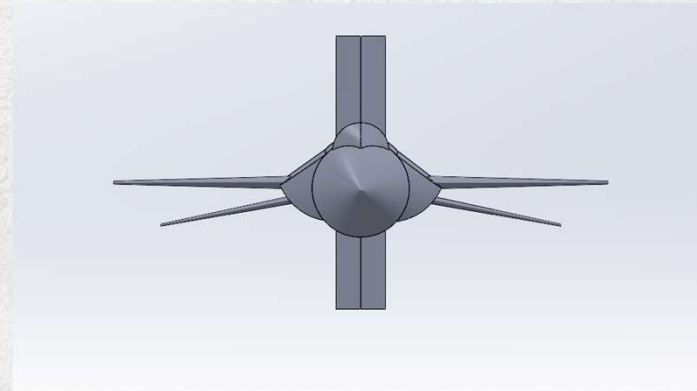
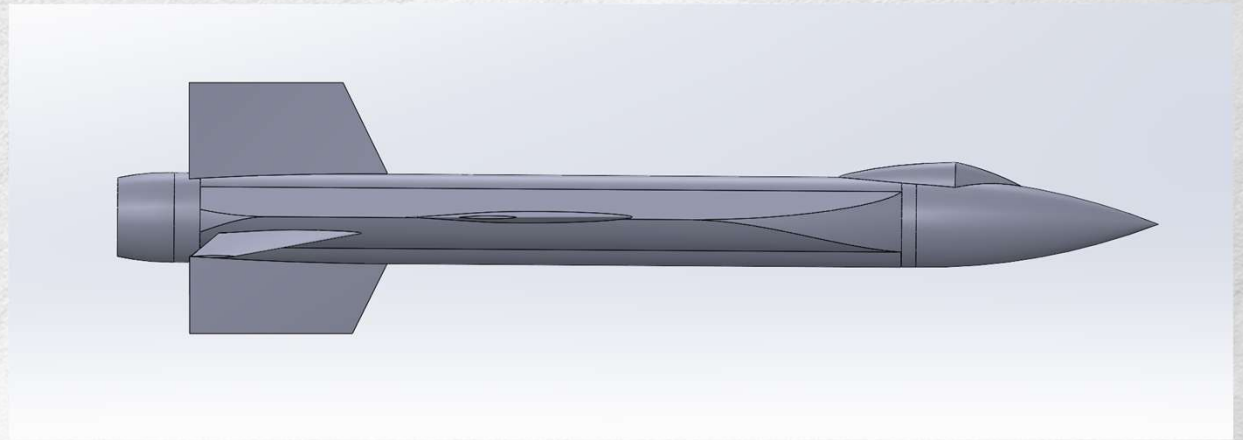
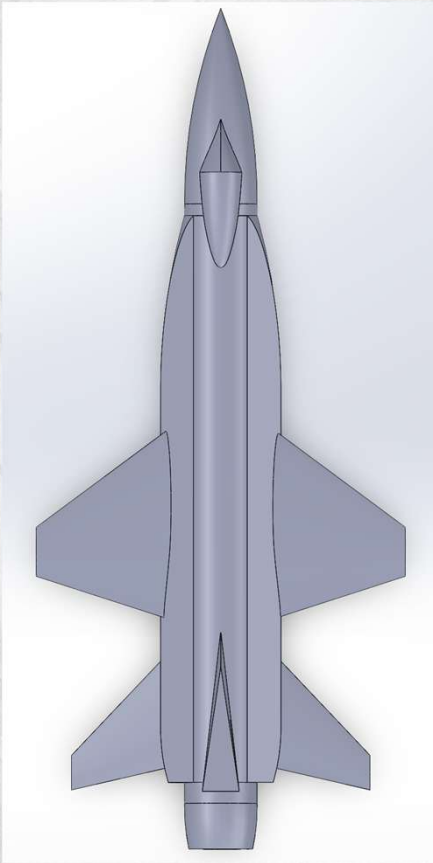


Ansys



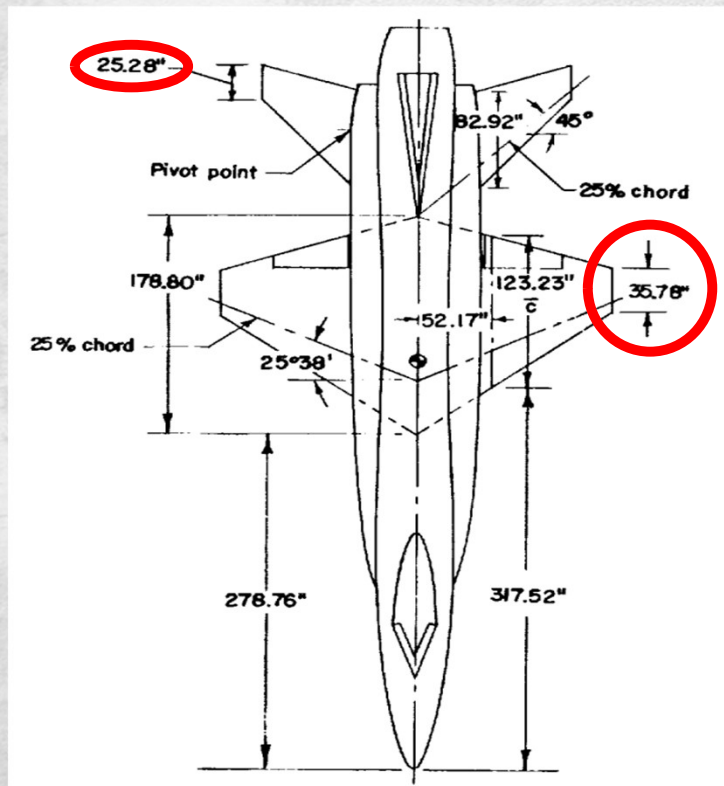
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Finalized SolidWorks Model

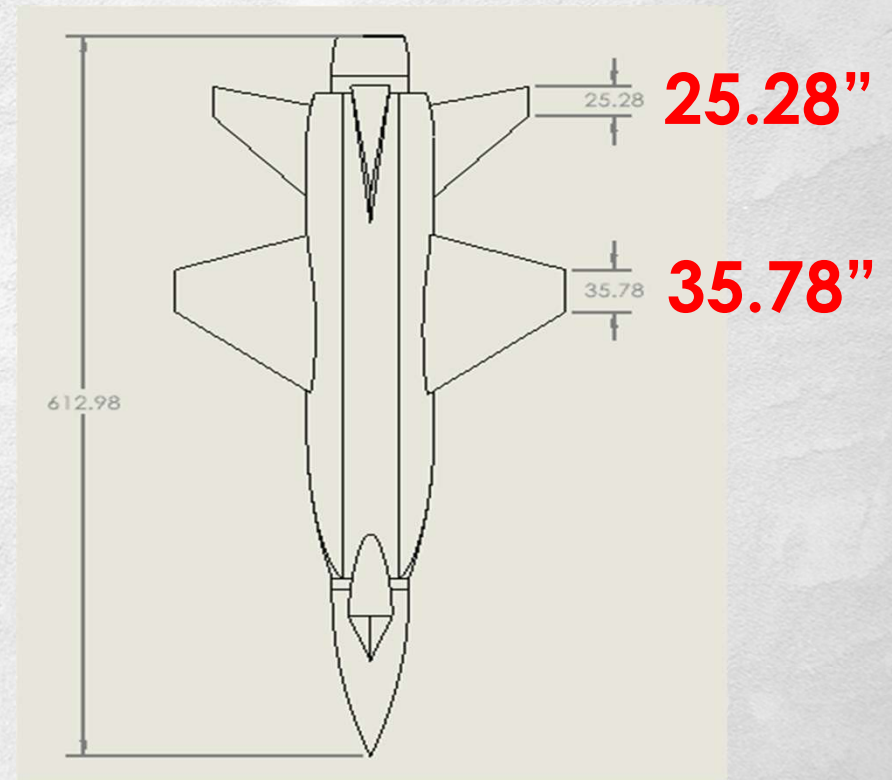


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Dimension Verification

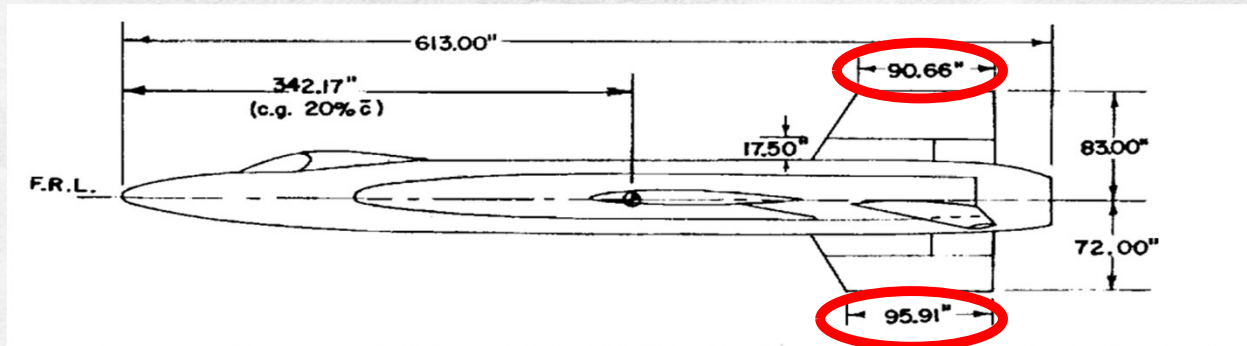


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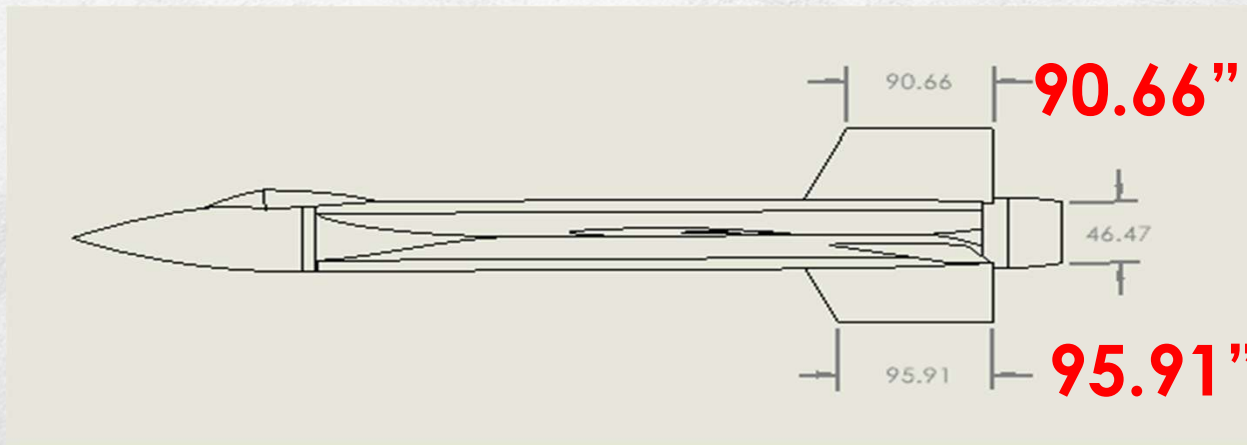


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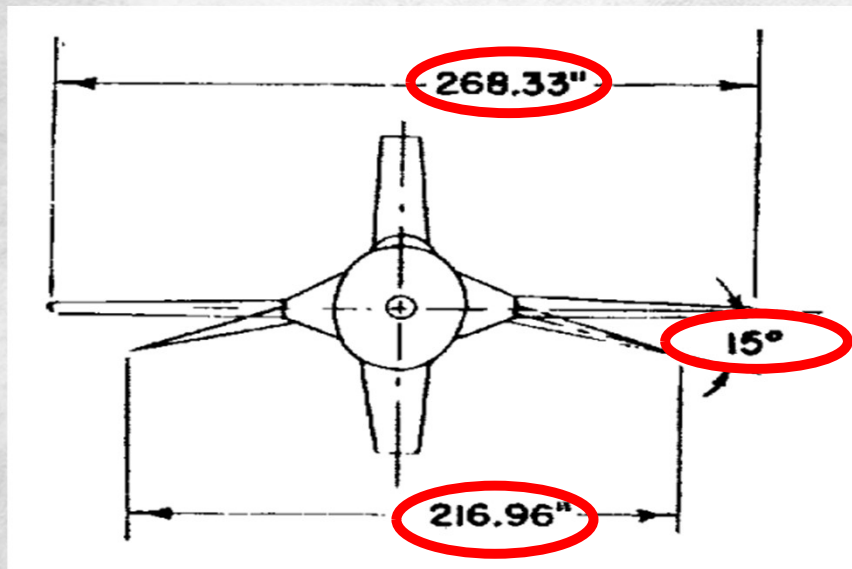
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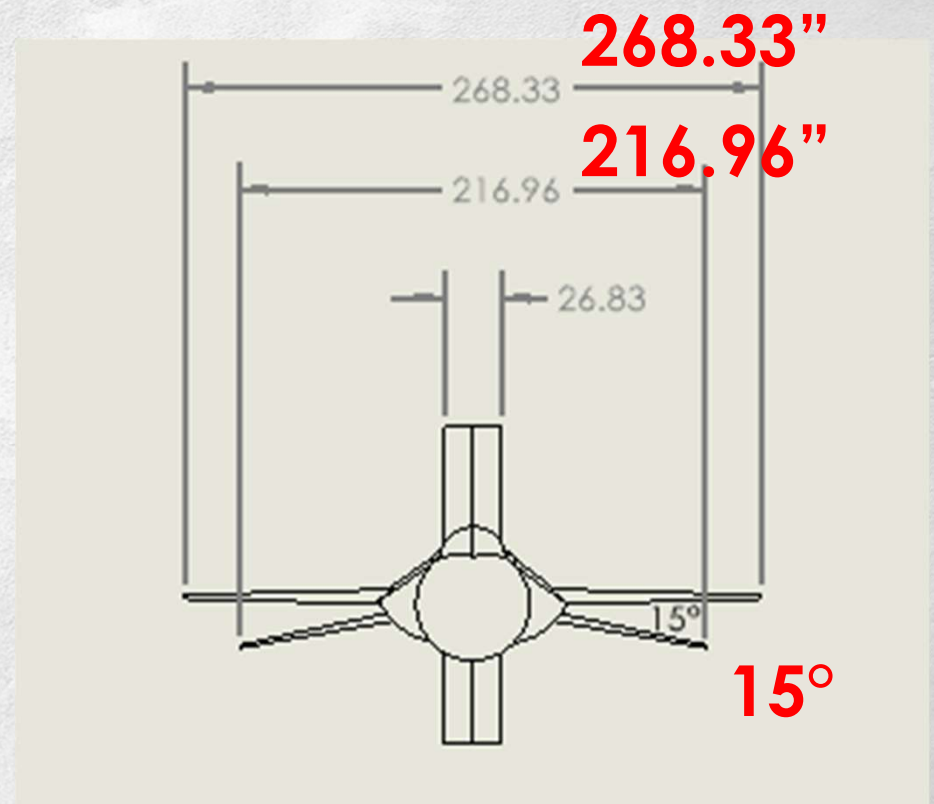
[3]



Dimension Verification



[3]



ANSYS Workflow

Geometry

- Defined spherical fluid domain
- Subtracted X-15 airframe from domain

Mesh

- Initialized mesh with default ANSYS settings
- Checked and refined mesh

Setup

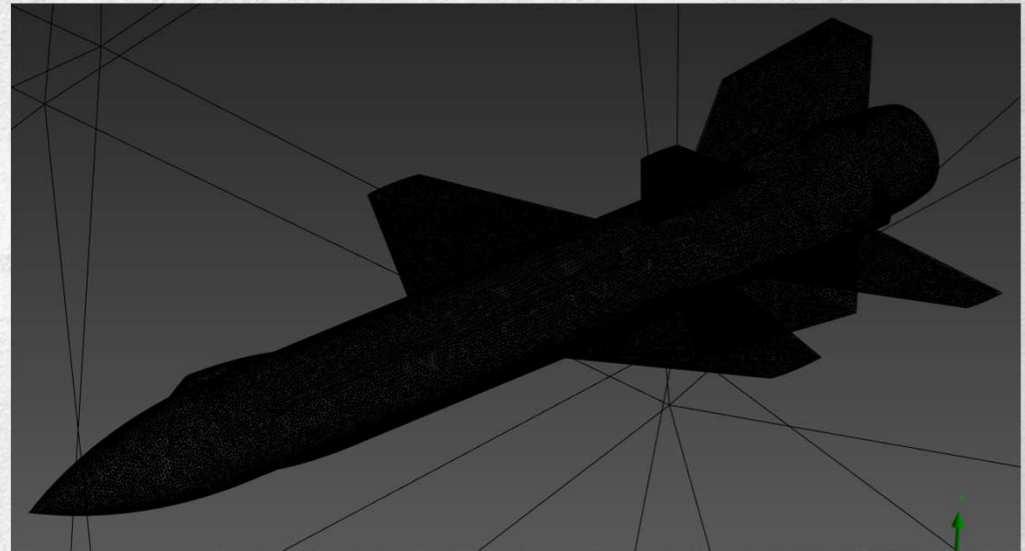
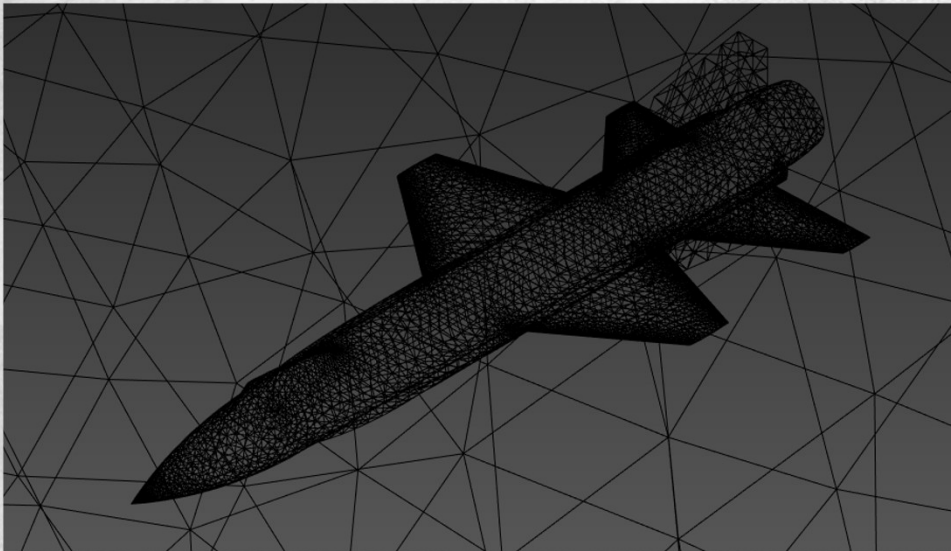
- Ran a time-steady, Density-based solver
- Enabled Energy equation and viscous model (k- ω SST)
- Input freestream conditions at desired altitude

Execution

- Initialize solution
- Run simulation
- Analyze results against technical report data using static pressure contours



ANSYS Meshing



Governing Equations

Gay-Lussac's Law

$$\frac{P_{\infty}}{T_{\infty}} = \frac{P_1}{T_1}$$

Fahrenheit to Rankin

$$R = F + 491.67$$

Example application at 100,00 feet

Freestream Pressure and Temperature at 100,000 feet:

$$\frac{P_{\infty}}{T_{\infty}} = \frac{23.328 \text{ psf}}{408.57 \text{ R}} = 0.0571$$

Solve for temperature using a known static pressure point on the airframe:

$$\frac{P_1}{0.0571} = T_1 \rightarrow \frac{125 \text{ psf}}{0.0571} = 2189.27 \text{ R}$$

$$2189.27 \text{ R} = F + 491.67 \rightarrow F = 1697.59^{\circ}\text{F}$$



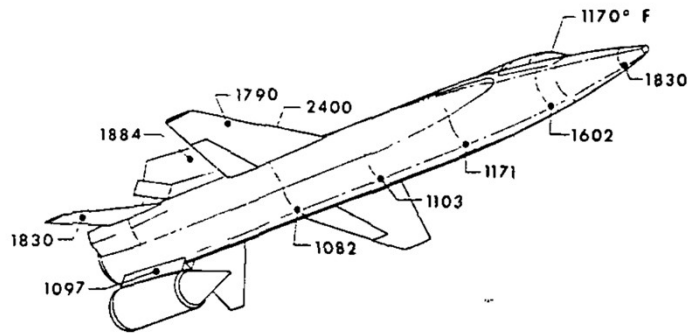
ANSYS Results

Mach: 8 / Altitude: 100,000 ft

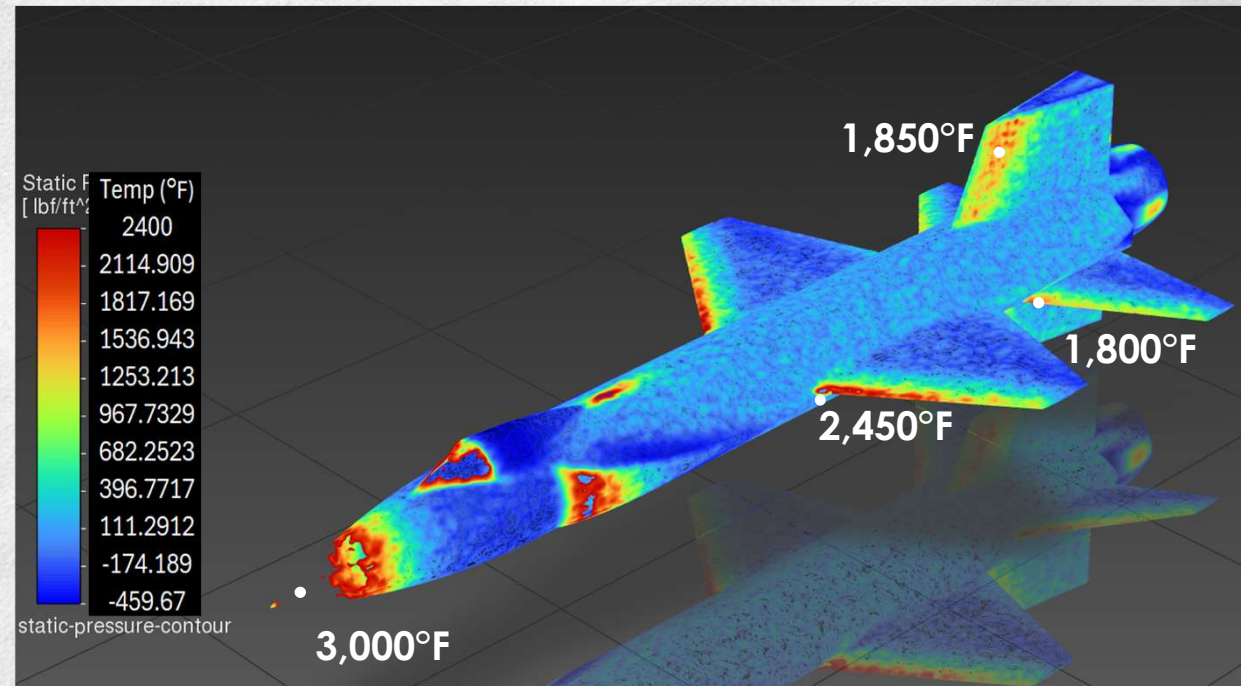
NASA's Analytical Results

CALCULATED TEMPERATURES ON UNPROTECTED X-15-2 AIRFRAME

MAXIMUM VELOCITY = 8,000 fps
ALTITUDE = 100,000 ft



[2]



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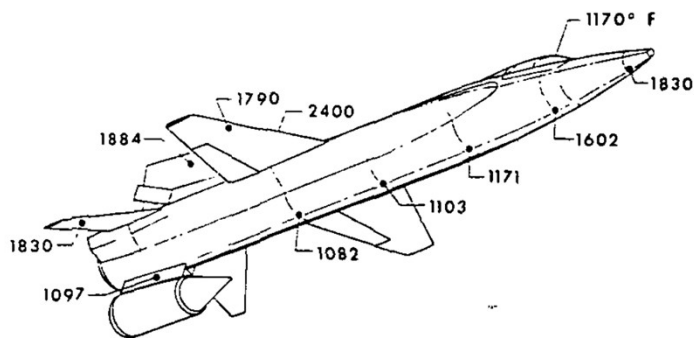
ANSYS Results

NASA's Analytical Results

Mach: 8 / Altitude: 100,000 ft

CALCULATED TEMPERATURES ON UNPROTECTED X-15-2 AIRFRAME

MAXIMUM VELOCITY = 8,000 fps
ALTITUDE = 100,000 ft



[2]



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ANSYS Verification

Appendix 9 X-15 Program Flight Log

	Flight No.	Flight ID	Serial No.	Date	Pilot	Max. Mach	Max. Altitude	Max. Speed
	44	1-24-40	56-6670	17 Oct 61	Walker	5.74	108,600	3,900
	45	2-21-37	56-6671	09 Nov 61	White	6.04	101,600	4,093
	46	3-1-2	56-6672	20 Dec 61	Armstrong	3.76	81,000	2,502
	47	1-25-44	56-6670	10 Jan 62	Petersen	0.97	44,750	645
	48	3-2-3	56-6672	17 Jan 62	Armstrong	5.51	133,500	3,765
	49	3-5-7	56-6672	05 Apr 62	Armstrong	4.12	180,000	2,850
	50	1-26-46	56-6670	19 Apr 62	Walker	5.69	154,000	3,866
Flight number 45 represented the first Mach 6 flight made by any manned aircraft.	51	3-4-8	56-6672	20 Apr 62	Armstrong	5.31	207,500	3,789
	52	1-27-48	56-6670	30 Apr 62	Walker	4.94	246,700	3,489
	53	2-22-40	56-6671	08 May 62	Rushworth	5.34	70,400	3,524
	54	1-28-49	56-6670	22 May 62	Rushworth	5.03	100,400	3,450
	55	2-23-43	56-6671	01 Jun 62	White	5.42	132,600	3,675
	56	1-29-50	56-6670	07 Jun 62	Walker	5.39	103,600	3,672
	57	3-5-9	56-6672	12 Jun 62	White	5.02	184,600	3,517
	58	3-6-10	56-6672	21 Jun 62	White	5.08	246,700	3,641
Flight number 46 was the first flight for the third X-15.	59	1-30-51	56-6670	27 Jun 62	Walker	5.92	123,700	4,104
	60	2-24-44	56-6671	29 Jun 62	McKay	4.95	83,200	3,280
	61	1-31-52	56-6670	16 Jul 62	Walker	5.37	107,200	3,674
	62	3-7-14	56-6672	17 Jul 62	White	5.45	314,750	3,832
	63	2-25-45	56-6671	19 Jul 62	McKay	5.18	85,250	3,474
	64	1-32-53	56-6670	26 Jul 62	Armstrong	5.74	98,900	3,989
	65	3-8-16	56-6672	02 Aug 62	Walker	5.07	144,500	3,438
Flight number 52 set an FAI certified altitude record.	66	2-26-46	56-6671	08 Aug 62	Rushworth	4.40	90,877	2,943
	67	3-9-18	56-6672	14 Aug 62	Walker	5.25	193,600	3,747
	68	2-27-47	56-6671	20 Aug 62	Rushworth	5.24	88,900	3,534
	69	2-28-48	56-6671	29 Aug 62	Rushworth	5.12	97,200	3,447
	70	2-29-50	56-6671	28 Sep 62	McKay	4.22	68,200	2,765
	71	3-10-19	56-6672	04 Oct 62	Rushworth	5.17	112,200	3,493
Flight number 53 was the first flight with a dynamic pressure over 2,000 psf.	72	2-30-51	56-6671	09 Oct 62	McKay	5.46	130,200	3,716
	73	3-11-20	56-6672	23 Oct 62	Rushworth	5.47	134,500	3,716
	74	2-31-52	56-6671	09 Nov 62	McKay	1.49	53,950	1,019
	75	3-12-22	56-6672	14 Dec 62	White	5.65	141,400	3,742
	76	3-13-23	56-6672	20 Dec 62	Walker	5.73	160,400	3,793
	77	3-14-24	56-6672	17 Jan 63	Walker	5.47	271,700	3,677
	78	1-33-54	56-6670	11 Apr 63	Rushworth	4.25	74,400	2,864
	79	3-15-25	56-6672	18 Apr 63	Walker	5.51	92,500	3,770
Flight number 62 set another FAI certified altitude record for class.	80	1-34-55	56-6670	25 Apr 63	McKay	5.32	105,500	3,654
	81	3-16-26	56-6672	02 May 63	Walker	4.73	209,400	3,488
	82	3-17-28	56-6672	14 May 63	Rushworth	5.20	95,600	3,600
	83	1-35-56	56-6670	15 May 63	McKay	5.57	124,200	3,856
	84	3-18-29	56-6672	29 May 63	Walker	5.52	92,000	3,838
	85	3-19-30	56-6672	18 Jun 63	Rushworth	4.97	223,700	3,539
	86	1-36-57	56-6670	25 Jun 63	Walker	5.51	111,800	3,911
	87	3-20-31	56-6672	27 Jun 63	Rushworth	4.89	285,000	3,425
Flight number 91 was the highest X-15 flight, 354,200 feet—almost 67 miles high.	88	1-37-59	56-6670	09 Jul 63	Walker	5.07	226,400	3,631
	89	1-38-61	56-6670	18 Jul 63	Rushworth	5.63	104,800	3,925
	90	3-21-32	56-6672	19 Jul 63	Walker	5.50	347,800	3,710
	91	3-22-36	56-6672	25 Aug 63	Walker	5.58	354,200	3,794
	92	1-39-63	56-6670	07 Oct 63	Eagle	4.21	77,800	2,834
	93	1-40-64	56-6670	29 Oct 63	Thompson	4.10	74,400	2,712
	94	3-23-39	56-6672	07 Nov 63	Rushworth	4.40	82,300	2,925
	95	1-41-65	56-6670	14 Nov 63	Eagle	4.75	90,800	3,286

Monographs in Aerospace History Number 18 — Hypersonics Before the Shuttle

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[4]

Flight No.	Flight ID	Serial No.	Date	Pilot	Max. Mach	Max. Altitude	Max. Speed
44	1-24-40	56-6670	17 Oct 61	Walker	5.74	108,600	3,900
45	2-21-37	56-6671	09 Nov 61	White	6.04	101,600	4,093
46	3-1-2	56-6672	20 Dec 61	Armstrong	3.76	81,000	2,502
47	1-25-44	56-6670	10 Jan 62	Petersen	0.97	44,750	645
48	3-2-3	56-6672	17 Jan 62	Armstrong	5.51	133,500	3,765

[4]



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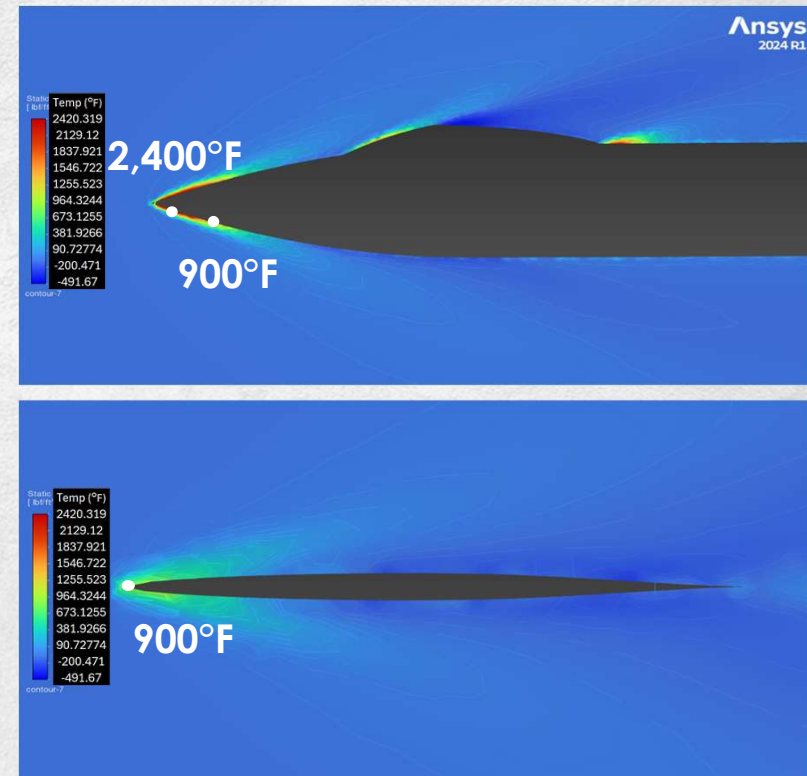
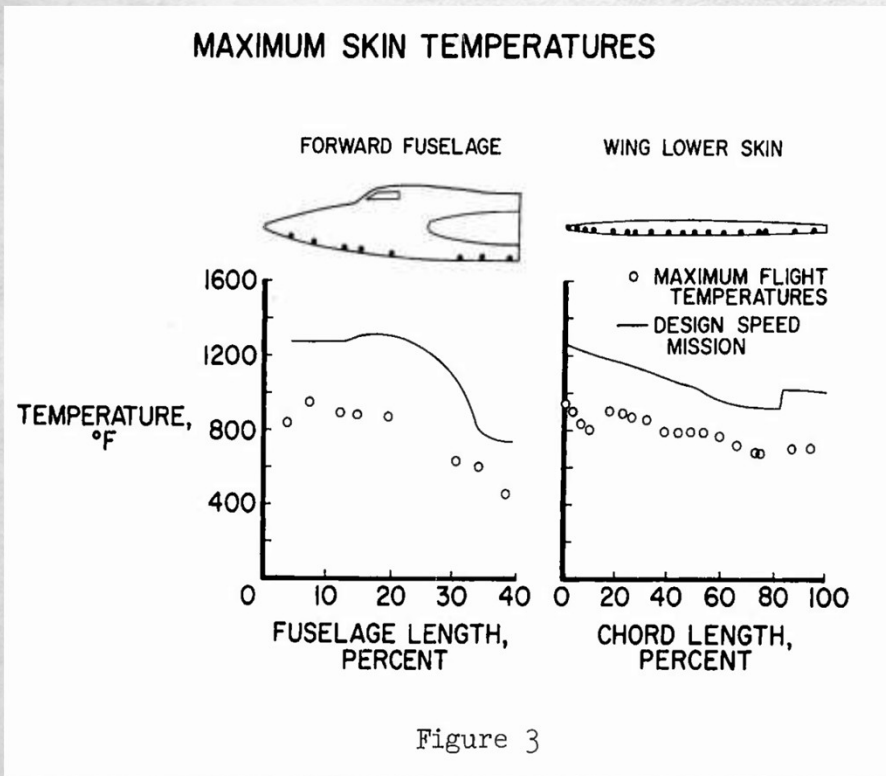
ANSYS Verification Results

NASA's Analytical Results

Mach: 6.04

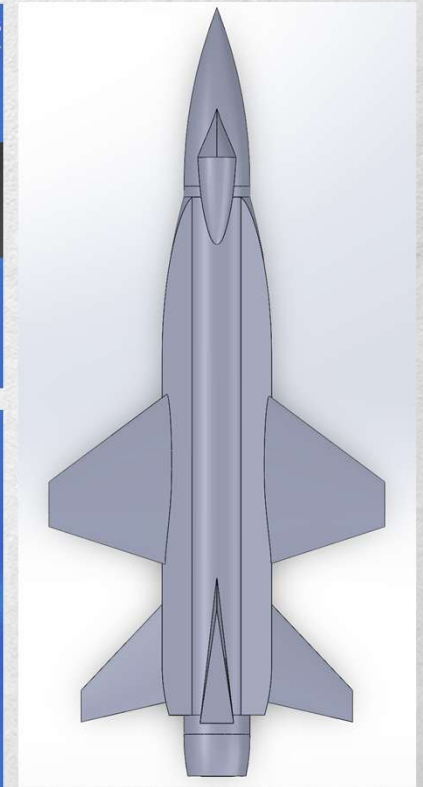
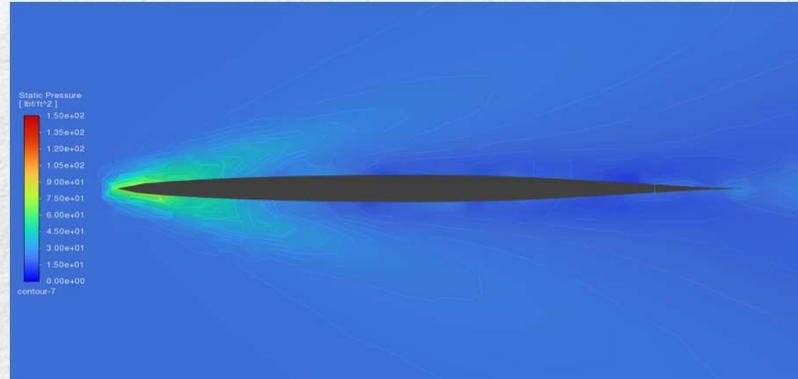
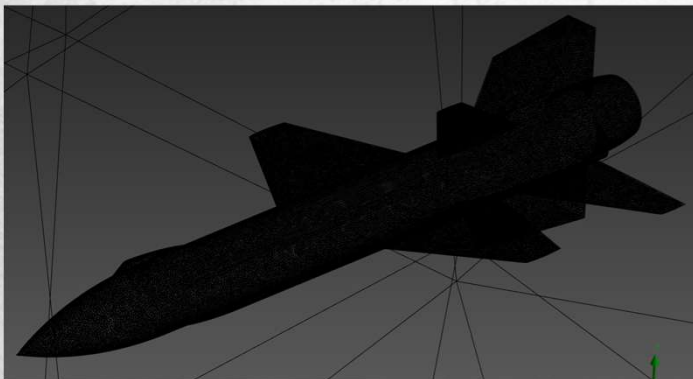
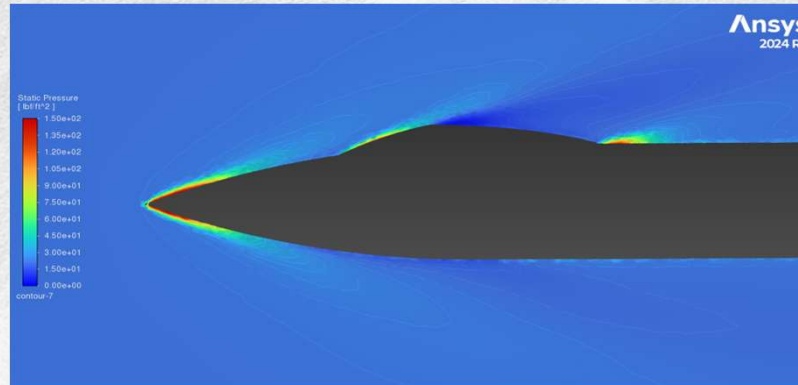
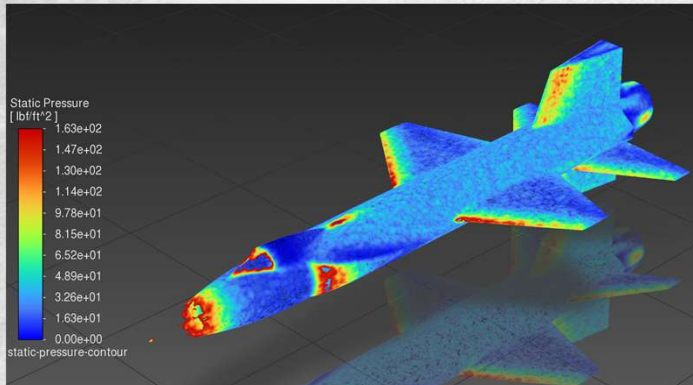
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Altitude: 101,600 ft



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Conclusion



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References

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- [2] NASA, "Progress of the X-15 Research Airplane Program," in Proceedings of the 4th Conference on Progress of the X-15 Research Airplane Program, Edwards AFB, CA, Oct. 7, 1965, NASA-SP-90.
- [3] W. D. Grantham and S. H. Scher, "Analytical investigation and prediction of spin and recovery characteristics of the North American X-15 airplane," NASA Technical Memorandum NASA-TM-X-294, Oct. 1, 1960.
- [4] D. R. Jenkins, *Hypersonics Before the Shuttle: A Concise History of the X-15 Research Airplane*, NASA History Series, Monographs in Aerospace History no. 18, NASA Publication SP-2000-4518, June 2000.
- [5] E. E. Kordes, "Structural Heating Experiences on the X-15 Airplane," NASA Technical Memorandum NASA-TM-X-711, Mar. 1962.

